DELETION FORMULAS

Abstract. Aloha

1. Introduction

2. Preliminaries

2.1. **Conventions.** For a finite matroid N, write $E(N)$ for its ground set, $\mathscr{L}(N)$ for its lattice of flats, and $\mathrm{rk}(N)$ and $\mathrm{crk}_N(A) = \mathrm{rk}(N) - \mathrm{rk}_N(A)$ for rank and corank. We use the standard minor notation $N|A$, N/A, and $N \setminus e$. If a and b are functions on finite matroids, their convolution is

$$(a * b)_N = \sum_{A \in \mathscr{L}(N)} a_{N|A} b_{N/A}.$$

For a polynomial-valued invariant a, its reversal is denoted by

$$a_N^{\mathrm{rev}}(x) = x^{\mathrm{rk}(N)} a_N(x^{-1}).$$

The symbols P, H, H^*, Q, Y, Z, F, and G^* denote the polynomial invariants introduced in the applications below; their required convolution identities are recorded there. Whenever an invariant is extended by zero to matroids with loops, that extension is stated explicitly at the point where it is used.

For a matroid M and an element $e \in E(M)$, define the boundary sets

$$S_e(M) = \{F \in \mathscr{L}(M) \mid e \notin F \text{ and } F \cup \{e\} \in \mathscr{L}(M)\},$$
$$\mathscr{T}_e(M) = \{B \in \mathscr{L}(M) \mid e \in B, B \setminus \{e\} \notin \mathscr{L}(M)\},$$

and put $\underline{S}_e(M) = S_e(M) \setminus \{\emptyset\}$. No non-coloop assumption is part of this notation; such an assumption is made only when required by a deletion formula.

We define τ by

$$\tau(M) := \begin{cases} [x^{\frac{\mathrm{rk}(M)-1}{2}}] P_M(x) & \text{if } \mathrm{rk}(M) \text{ is odd,} \\ 0 & \text{if } \mathrm{rk}(M) \text{ is even.} \end{cases}$$

2.2. **Flat calculus.** We recall the deletion formula for the characteristic polynomial:

Lemma 2.1. *Let N be a matroid and $e \in E(N)$. If e is a coloop of N, then*

$$\chi_N(x) = (x-1)\chi_{N/e}(x).$$

If e is not a coloop of N, then

$$\chi_N(x) = \chi_{N\setminus e}(x) - \chi_{N/e}(x).$$

The following two lemmas collect the flat decompositions and convolution reindexings used throughout the deletion formulas.

Lemma 2.2 (Flat calculus at a non-coloop)**.** *Let M be loopless and let $e \in E(M)$ be a non-coloop. Put*

$$\mathscr{A}_e(M) = \{A \in \mathscr{L}(M) : e \notin A\}, \qquad S = S_e(M), \qquad T = \mathscr{T}_e(M).$$

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Then

$$\mathscr{L}(M) = \mathscr{A}_e(M) \,\dot\cup\, \{A \cup \{e\} : A \in S\} \,\dot\cup\, T,$$
$$\mathscr{L}(M \setminus e) = \mathscr{A}_e(M) \,\dot\cup\, \{B \setminus \{e\} : B \in T\},$$
$$\mathscr{L}(M/e) = S \,\dot\cup\, \{B \setminus \{e\} : B \in T\}.$$

For every $A \in \mathscr{A}_e(M)$, the matroid M/A is loopless, e is not a coloop of M/A, and the maps $K \mapsto A \cup K$ identify

$$S_e(M/A) = \{F \setminus A : F \in S,\ A \subseteq F\},$$
$$\mathscr{T}_e(M/A) = \{F \setminus A : F \in T,\ A \subseteq F\}.$$

For every flat $B \in \mathscr{L}(M)$ containing e,

$$S_e(M|B) = \{F \in S : F \subseteq B\},$$

and, if $B \in T$, then

$$\mathscr{T}_e(M|B) = \{K \in T : K \subseteq B\}.$$

Finally, the map $(F, B) \mapsto (F, B \setminus F)$ is a bijection

$$\big\{(F, B) : B \in T,\ F \in S_e(M|B)\big\} \longrightarrow \big\{(F, K) : F \in S,\ K \in \mathscr{T}_e(M/F)\big\}.$$

Proof. If a flat of M contains e, then either removing e leaves a flat, which gives a unique member of S, or it does not, which gives a member of T. This proves the first decomposition. If $C \in \mathscr{L}(M \setminus e)$, then $\mathrm{cl}_M(C)$ is either C or $C \cup \{e\}$. In the former case $C \in \mathscr{A}_e(M)$, while in the latter $C = B \setminus \{e\}$ for the unique $B = \mathrm{cl}_M(C) \in T$; the converse statements follow in the same way. This proves the second decomposition. Finally,

$$\mathrm{cl}_{M/e}(C) = \mathrm{cl}_M(C \cup \{e\}) \setminus \{e\} \qquad \big(C \subseteq E(M) \setminus \{e\}\big).$$

Thus, if $C \in \mathscr{L}(M/e)$ and $B = \mathrm{cl}_M(C \cup \{e\})$, then either $B \setminus \{e\} = C$ is a flat of M, so $C \in S$, or $B \in T$ and $C = B \setminus \{e\}$. Conversely, every flat in either displayed class is a flat of M/e. This proves the third decomposition. If $A \in \mathscr{A}_e(M)$, then M/A is loopless because A is a flat, and

$$\mathrm{rk}\big((M/A) \setminus e\big) = \mathrm{rk}(M \setminus e) - \mathrm{rk}(A) = \mathrm{rk}(M) - \mathrm{rk}(A) = \mathrm{rk}(M/A),$$

so e is not a coloop of M/A. The stated descriptions of its boundary sets follow from the interval isomorphism between the flats of M/A and the flats of M containing A. Likewise, the flats of $M|B$ are the flats of M contained in B, which proves the two restriction identities. The last bijection has inverse $(F, K) \mapsto (F, F \cup K)$; the flat descriptions just proved show that these maps have the claimed domains. □

Lemma 2.3 (Convolution reindexing)**.** *In the setting of Lemma 2.2, let u and v be functions on finite matroids and put*

$$w_N = \sum_{A \in \mathscr{L}(N)} u_{N|A} v_{N/A}.$$

Then

$$w_{M/e} = \sum_{A \in S} u_{M|A} v_{M/(A \cup \{e\})} + \sum_{B \in T} u_{(M|B)/e} v_{M/B}, \tag{1}$$

$$w_{M|F} = \sum_{\substack{A \in \mathscr{L}(M) \\ A \subseteq F}} u_{M|A} v_{(M|F)/A} \qquad (F \in S), \tag{2}$$

$$w_{M/(L \cup \{e\})} = \sum_{\substack{B \in \mathscr{L}(M) \\ B \supseteq L \cup \{e\}}} u_{(M|B)/(L \cup \{e\})} v_{M/B} \qquad (L \in S), \tag{3}$$

$$w_{M/K} = \sum_{\substack{B\in\mathcal{L}(M)\\ B\supseteq K}} u_{(M|B)/K} v_{M/B} \qquad (K \in T). \tag{4}$$

Proof. For the first identity, the third decomposition in Lemma 2.2 applies, and the corresponding minors are

$$(M/e)|A = M|A, \quad (M/e)/A = M/(A \cup \{e\}) \quad (A \in S),$$

and

$$(M/e)|(B \setminus \{e\}) = (M|B)/e, \quad (M/e)/(B \setminus \{e\}) = M/B \quad (B \in T).$$

For the remaining identities, use the correspondence between flats of M/H and flats B of M containing H, together with

$$(M/H)|(B \setminus H) = (M|B)/H, \qquad (M/H)/(B \setminus H) = M/B.$$

Taking $H = F$, $L \cup \{e\}$, and K, respectively, proves the claim. □

3. Convolution Deletion Formula

The next theorem records the bookkeeping needed to pass a pair of deletion recurrences through a convolution. It retains the terms coming from contractions by nonflats, which vanish in the loop-vanishing specializations used below.

Theorem 3.1 (Convolution Deletion Formula)**.** *Let R be a commutative ring, and let $g, h, \Phi_h, \Psi_h, \Phi_g$, and Ψ_g be isomorphism-invariant functions on finite matroids with values in R. Define*

$$f_N = \sum_{A\in\mathcal{L}(N)} g_{N|A} h_{N/A}.$$

Suppose that there are fixed scalars $\alpha_h, \alpha_g, \gamma_g \in R$ such that, for every pointed loopless matroid (N, p) with p not a coloop, one has

$$h_N = h_{N\setminus p} + \alpha_h h_{N/p} + \sum_{K_h\in S_p(N)} \Phi_h\big(N/(K_h \cup \{p\})\big) h_{N|K_h} + \sum_{L_h\in\mathcal{T}_p(N)} \Psi_h(N/L_h) h_{(N|L_h)/p},$$

$$g_N = g_{N\setminus p} + \alpha_g g_{N/p} + \sum_{K_g\in S_p(N)} \Phi_g(N|K_g) g_{N/(K_g\cup\{p\})} + \sum_{L_g\in\mathcal{T}_p(N)} \Psi_g\big((N|L_g)/p\big) g_{N/L_g},$$

and that, whenever p is a coloop of a loopless matroid N,

$$g_N = \gamma_g g_{N/p} + \sum_{K_g\in S_p(N)} \Phi_g(N|K_g) g_{N/(K_g\cup\{p\})}.$$

Let M be loopless and let $e \in E(M)$ not be a coloop. With

$$\delta = \alpha_h + \gamma_g - \alpha_g,$$

the following identity holds:

$$\begin{aligned} f_M = f_{M\setminus e} &+ \alpha_g f_{M/e} + \delta \sum_{A\in S_e(M)} g_{M|A} h_{M/(A\cup\{e\})} \\ &+ \alpha_h \sum_{\substack{A\in\mathcal{L}(M),\ e\notin A\\ A\cup\{e\}\notin\mathcal{L}(M)}} g_{M|A} h_{M/(A\cup\{e\})} \\ &+ \sum_{F\in S_e(M)} \Phi_h\big(M/(F\cup\{e\})\big) f_{M|F} \\ &+ \sum_{F\in\mathcal{T}_e(M)} \Psi_h(M/F) \sum_{\substack{A\in\mathcal{L}(M|F)\\ e\notin A}} g_{(M|F)|A} h_{(M|F)/(A\cup\{e\})} \end{aligned}$$

$$+ \sum_{L \in S_e(M)} \Phi_g(M|L) f_{M/(L\cup\{e\})}$$
$$+ \sum_{K \in \mathcal{T}_e(M)} \Psi_g\big((M|K)/e\big) f_{M/K}.$$

Proof. Put

$$\mathcal{A} = \{A \in \mathcal{L}(M) : e \notin A\}, \qquad S = S_e(M), \qquad T = \mathcal{T}_e(M).$$

Lemma 2.2 gives all flat decompositions used below. In particular, for every $A \in \mathcal{A}$, the matroid M/A is loopless and e is not a coloop of M/A. Set

$$X = \sum_{A \in \mathcal{A}} g_{M|A} h_{M/A}.$$

By the boundary-set description in Lemma 2.2, the recurrence for each $h_{M/A}$ is indexed by the flats $F \in S \cup T$ containing A. The relevant minor identities are

$$(M/A)/(K \cup \{e\}) = M/(F \cup \{e\}), \quad (M/A)|K = (M|F)/A,$$

and

$$(M/A)/J = M/F, \quad ((M/A)|J)/e = ((M|F)/A)/e,$$

where $K = F \setminus A$ and $J = F \setminus A$. Thus, after multiplying by $g_{M|A}$ and summing over $A \in \mathcal{A}$, we group the deletion, contraction, Φ_h, and Ψ_h terms separately. Lemma 2.2 and the restriction reindexing identity (2) give

$$X = \underbrace{f_{M\setminus e} - \sum_{B \in T} g_{M|(B\setminus\{e\})} h_{M/B}}_{\text{deletion terms}} \tag{5}$$
$$+ \underbrace{\alpha_h \sum_{A \in S} g_{M|A} h_{M/(A\cup\{e\})}}_{\text{contraction terms with } A\cup\{e\} \text{ flat}} \tag{6}$$
$$+ \alpha_h \sum_{\substack{A \in \mathcal{L}(M),\ e \notin A \\ A\cup\{e\} \notin \mathcal{L}(M)}} g_{M|A} h_{M/(A\cup\{e\})} \tag{7}$$
$$+ \underbrace{\sum_{F \in S} \Phi_h\big(M/(F \cup \{e\})\big) f_{M|F}}_{\Phi_h\text{-boundary terms}} \tag{8}$$
$$+ \sum_{F \in T} \Psi_h(M/F) \sum_{\substack{A \in \mathcal{L}(M|F) \\ e \notin A}} g_{(M|F)|A} h_{(M|F)/(A\cup\{e\})}. \tag{9}$$

The Φ_h- and Ψ_h-boundary terms are obtained by exchanging the sums over A and F; for $F \in S$, the resulting inner convolution is $f_{M|F}$, by (2). Here the two contraction terms (6) and (7) are separated according as $A \cup \{e\}$ is, or is not, a flat. The deletion term follows from the deletion-flat decomposition in Lemma 2.2: if $B \in T$, then e is a loop of $M/(B \setminus \{e\})$, and hence $(M \setminus e)/(B \setminus \{e\}) = M/B$.

Put

$$Y = \sum_{A \in S} g_{M|(A\cup\{e\})} h_{M/(A\cup\{e\})} + \sum_{B \in T} g_{M|B} h_{M/B}.$$

Then $f_M = X + Y$. If $A \in S$, then $\mathrm{rk}(A \cup \{e\}) = \mathrm{rk}(A) + 1$, so e is a coloop of $M|(A \cup \{e\})$ and $(M|(A \cup \{e\}))/e = M|A$. If $B \in T$, then $e \in \mathrm{cl}_M(B \setminus \{e\})$, so e is not a coloop of $M|B$.

Moreover,

$$S_e(M|(A \cup \{e\})) = \{L \in S : L \subseteq A\},$$

and, for $B \in T$,

$$S_e(M|B) = \{L \in S : L \subseteq B\}, \qquad \mathcal{T}_e(M|B) = \{K \in T : K \subseteq B\}.$$

Apply the same grouping argument to Y, now using the recurrence for g, the preceding local boundary identities from Lemma 2.2, and (3)–(4). This yields

$$\begin{aligned} Y = &\sum_{B\in T} g_{M|(B\setminus\{e\})} h_{M/B} \\ &+ \gamma_g \sum_{A\in S} g_{M|A} h_{M/(A\cup\{e\})} + \alpha_g \sum_{B\in T} g_{(M|B)/e} h_{M/B} \\ &+ \sum_{L\in S} \Phi_g(M|L) f_{M/(L\cup\{e\})} \\ &+ \sum_{K\in T} \Psi_g\big((M|K)/e\big) f_{M/K}. \end{aligned} \tag{10}$$

Finally, (1) gives

$$f_{M/e} = \sum_{A\in S} g_{M|A} h_{M/(A\cup\{e\})} + \sum_{B\in T} g_{(M|B)/e} h_{M/B}.$$

Adding (9) and (10), the two sums indexed by T with factor $g_{M|(B\setminus\{e\})} h_{M/B}$ cancel. The displayed identity for $f_{M/e}$ and the equality $\delta = \alpha_h + \gamma_g - \alpha_g$ then transform the remaining scalar terms into the first two lines of the asserted formula. The four boundary sums are already the remaining four lines of that formula. □

Corollary 3.2 (Loop-vanishing convolution deletion formula)**.** *In the setting of Theorem 3.1, additionally assume that $h_N = 0$ whenever N has a loop, that $\Psi_h = 0$, and that $\Phi_g(\emptyset) = 0$. Then*

$$\begin{aligned} f_M = f_{M\setminus e} + \alpha_g f_{M/e} + \delta &\sum_{A\in S_e(M)} g_{M|A} h_{M/(A\cup\{e\})} \\ &+ \sum_{F\in S_e(M)} \Phi_h\big(M/(F\cup\{e\})\big) f_{M|F} \\ &+ \sum_{L\in S_e(M)} \Phi_g(M|L) f_{M/(L\cup\{e\})} \\ &+ \sum_{K\in \mathcal{T}_e(M)} \Psi_g\big((M|K)/e\big) f_{M/K}. \end{aligned}$$

Proof. Let $A \in \mathcal{L}(M)$ avoid e and suppose that $A \cup \{e\}$ is not a flat. Choose

$$z \in \mathrm{cl}_M(A \cup \{e\}) \setminus (A \cup \{e\}).$$

Then

$$\mathrm{rk}_{M/(A\cup\{e\})}(\{z\}) = \mathrm{rk}_M(A \cup \{e, z\}) - \mathrm{rk}_M(A \cup \{e\}) = 0,$$

so z is a loop of $M/(A \cup \{e\})$. The loop-vanishing hypothesis therefore makes every summand in the nonflat-boundary sum zero. The hypothesis $\Psi_h = 0$ eliminates the Ψ_h-sum, and $\Phi_g(\emptyset) = 0$ makes the $L = \emptyset$ summand vanish. The theorem now reduces exactly to the displayed formula. □

Remark 3.3. The loop-vanishing hypothesis in the corollary cannot be omitted: for example, $h_N = 2^{|E(N)|}$ and $g_N = 1$, with $\alpha_h = \gamma_g = 1$, $\alpha_g = 0$, and all boundary weights zero, satisfy the recurrences but the nonflat-boundary sum for $\mathsf{U}_{2,3}$ is nonzero.

4. Inverse Deletion Formula

The next result is a companion to the Convolution Deletion Formula. It makes precise how a deletion recurrence for a convolution kernel transfers to its inverse. The residual term records exactly what can remain when contractions by nonflats create loops.

Theorem 4.1 (Inverse Deletion Formula)**.** *Let R be a commutative ring, and let κ, λ, and ϕ be isomorphism-invariant functions on finite matroids with values in R. Suppose that*

$$\kappa_\emptyset = \sigma \in \{1, -1\}$$

and that λ is a right convolution inverse of κ, namely

$$\sum_{F \in \mathscr{L}(N)} \kappa_{N|F} \lambda_{N/F} = \delta_{\emptyset,N} \qquad \text{for every finite matroid } N. \tag{11}$$

Assume that there are fixed scalars $\alpha, \gamma, \varepsilon \in R$ such that, for every pointed loopless matroid (N, p), one has

$$\kappa_N = \kappa_{N\setminus p} + \alpha \kappa_{N/p} + \sum_{F \in S_p(N)} \phi\big(N/(F \cup \{p\})\big) \kappa_{N|F} \qquad \text{if } p \text{ is not a coloop,} \tag{12}$$

$$\kappa_N = \gamma \kappa_{N/p} + \varepsilon \mathbf{1}_{\{N \cong \mathsf{U}_{1,1}\}} \qquad \text{if } p \text{ is a coloop.} \tag{13}$$

Then, for every loopless matroid M and every non-coloop $e \in E(M)$,

$$\begin{aligned} \lambda_M = \lambda_{M\setminus e} &+ \sigma(\alpha - \gamma) \sum_{A \in S_e(M)} \kappa_{M|A} \lambda_{M/(A \cup \{e\})} - \sigma \varepsilon \mathbf{1}_{\{\emptyset \in S_e(M)\}} \lambda_{M/e} \\ &- \sigma \sum_{\substack{A \in \mathscr{L}(M) \\ e \notin A,\ A \neq \emptyset}} \kappa_{M|A} \left(\lambda_{M/A} - \lambda_{(M/A)\setminus e} \right) \\ &- \sigma \sum_{A \in S_e(M)} \kappa_{M|A} \sum_{B \in \mathscr{T}_e(M/A)} \phi\big(((M/A)|B)/e\big) \lambda_{(M/A)/B}. \end{aligned} \tag{14}$$

Proof. Put

$$\mathscr{A} = \{A \in \mathscr{L}(M) : e \notin A\}, \qquad S = S_e(M), \qquad T = \mathscr{T}_e(M), \qquad \eta = \mathbf{1}_{\{\emptyset \in S\}}.$$

First suppose that $\sigma = 1$. We use Lemma 2.2. In particular, a flat of M containing e is either $A \cup \{e\}$ for a unique $A \in S$, or belongs to T.

Let

$$X = \sum_{\substack{B \in \mathscr{L}(M) \\ e \in B}} \kappa_{M|B} \lambda_{M/B}.$$

For $A \in S$, the element e is a coloop of $M|(A \cup \{e\})$, whereas for $B \in T$ it is not a coloop of $M|B$. Applying (13) in the first case and (12) in the second gives

$$\begin{aligned} X = \gamma &\sum_{A \in S} \kappa_{M|A} \lambda_{M/(A \cup \{e\})} + \sum_{B \in T} \kappa_{M|(B\setminus\{e\})} \lambda_{M/B} \\ &+ \alpha \sum_{B \in T} \kappa_{(M|B)/e} \lambda_{M/B} + \sum_{A \in S} \kappa_{M|A} \sum_{C \in \mathscr{T}_e(M/A)} \phi\big(((M/A)|C)/e\big) \lambda_{(M/A)/C} + \eta \varepsilon \lambda_{M/e}. \end{aligned} \tag{15}$$

Indeed, the exceptional summand in the coloop recurrence occurs only when $A = \emptyset$, and this index is present precisely when $\emptyset \in S$. For the last term, we have interchanged the two boundary sums using the final bijection of Lemma 2.2. Under this bijection,

$$\frac{M|B}{F \cup \{e\}} = \frac{(M/F)|K}{e}, \qquad \frac{M}{B} = \frac{M/F}{K}.$$

Since e is not a coloop, M/e has nonempty ground set, so applying (11) to M/e and using (1) yields

$$0 = \sum_{A\in S} \kappa_{M|A}\lambda_{M/(A\cup\{e\})} + \sum_{B\in T} \kappa_{(M|B)/e}\lambda_{M/B}. \tag{16}$$

Substituting (16) into (15), we obtain

$$X = \sum_{B\in T} \kappa_{M|(B\setminus\{e\})}\lambda_{M/B} + (\gamma-\alpha)\sum_{A\in S} \kappa_{M|A}\lambda_{M/(A\cup\{e\})} + \sum_{A\in S} \kappa_{M|A} \sum_{C\in\mathcal{T}_e(M/A)} \phi\big(((M/A)|C)/e\big)\lambda_{(M/A)/C} + \eta\varepsilon\lambda_{M/e}. \tag{17}$$

The convolution identities for M and $M\setminus e$, respectively, are

$$0 = \lambda_M + \sum_{\substack{A\in\mathcal{A}\\ A\neq\emptyset}} \kappa_{M|A}\lambda_{M/A} + X,$$

$$0 = \lambda_{M\setminus e} + \sum_{\substack{A\in\mathcal{A}\\ A\neq\emptyset}} \kappa_{M|A}\lambda_{(M\setminus e)/A} + \sum_{B\in T} \kappa_{M|(B\setminus\{e\})}\lambda_{M/B}.$$

Eliminating the common T-sum with (17) gives

$$\lambda_M = \lambda_{M\setminus e} + (\alpha-\gamma)\sum_{A\in S} \kappa_{M|A}\lambda_{M/(A\cup\{e\})} - \sum_{A\in S} \kappa_{M|A} \sum_{C\in\mathcal{T}_e(M/A)} \phi\big(((M/A)|C)/e\big)\lambda_{(M/A)/C} - \sum_{\substack{A\in\mathcal{A}\\ A\neq\emptyset}} \kappa_{M|A}\left(\lambda_{M/A} - \lambda_{(M\setminus e)/A}\right) - \eta\varepsilon\lambda_{M/e}. \tag{18}$$

Equation (18) is exactly (14) when $\sigma = 1$.

If $\sigma = -1$, put

$$\widetilde{\kappa} = -\kappa, \qquad \widetilde{\lambda} = -\lambda, \qquad \widetilde{\varepsilon} = -\varepsilon.$$

Then $\widetilde{\kappa}_\emptyset = 1$, $\widetilde{\lambda}$ is a right convolution inverse of $\widetilde{\kappa}$, and the two kernel recurrences hold for $\widetilde{\kappa}$ with the same α, γ, and ϕ, and with exceptional scalar $\widetilde{\varepsilon}$. Applying the case just proved and substituting back gives the displayed $\sigma = -1$ identity, equivalently (14). □

Corollary 4.2 (Loop-vanishing inverse deletion formula)**.** *In the setting of Theorem 4.1, additionally suppose that*

$$\varepsilon = 0 \qquad \textit{and} \qquad \lambda_N = \phi(N) = 0 \qquad \textit{whenever } N \textit{ has a loop.}$$

Then

$$\lambda_M = \lambda_{M\setminus e} + (\alpha-\gamma)\lambda_{M/e} - \sum_{B\in\mathcal{T}_e(M)} \phi\big((M|B)/e\big)\lambda_{M/B}. \tag{19}$$

Proof. Put

$$\mathcal{A} = \{A \in \mathcal{L}(M) : e \notin A\}, \qquad S = S_e(M).$$

We argue by induction on $|E(M)|$. Let $A \in \mathcal{A}\setminus S$ and put $N = M/A$. The matroid N is loopless, but $\{e\}$ is not a flat of N. Thus $N/e = M/(A\cup\{e\})$ has a loop, so its λ-term vanishes. Choose $z \in \mathrm{cl}_N(\{e\})\setminus\{e\}$. Every flat $B \in \mathcal{T}_e(N)$ contains $\mathrm{cl}_N(\{e\})$, hence contains z, and z is a loop of $(N|B)/e$. Thus every corresponding ϕ-term vanishes as well.

After grouping the terms indexed by S and using the reindexing bijection of Lemma 2.2, (14) becomes

$$\lambda_M - \lambda_{M\setminus e} - (\alpha-\gamma)\lambda_{M/e} + \sum_{B\in\mathcal{T}_e(M)} \phi\big((M|B)/e\big)\lambda_{M/B}$$

$$= -\sigma \sum_{\substack{A\in\mathcal{A}\\ A\neq\emptyset}} \kappa_{M|A}\left(\lambda_{M/A} - \lambda_{(M/A)\setminus e} - (\alpha-\gamma)\lambda_{(M/A)/e} + \sum_{C\in\mathcal{T}_e(M/A)} \phi\big(((M/A)|C)/e\big)\lambda_{(M/A)/C}\right).$$

Each parenthesized expression is the left-hand side of (19) for M/A. Since M/A is loopless, e is not a coloop in it, and $A \neq \emptyset$, it vanishes by induction. This proves (19). □

5. Deletion Formulas

5.1. **The Kazhdan-Lusztig Polynomial.**

Theorem 5.1 (Kazhdan–Lusztig deletion formula)**.** *Let M be loopless and let $e \in E(M)$ be a non-coloop. Then*

$$P_M(x) = P_{M\setminus e}(x) - xP_{M/e}(x) + \sum_{F\in S_e(M)} \tau\big(M/(F\cup\{e\})\big)x^{\mathrm{crk}_M(F)/2}P_{M|F}(x).$$

Proof. Apply the three flat decompositions of Lemma 2.2 to the convolution identity $P^{\mathrm{rev}}_M(x) = \sum_{B\in\mathcal{L}(M)} \chi_{M|B}(x)P_{M/B}(x)$. Using Lemma 2.1 on the factors indexed by flats containing e gives

$$P^{\mathrm{rev}}_M(x) = \sum_{A\not\ni e} \chi_{M|A}P_{M/A} + \sum_{A\in S_e(M)} (x-1)\chi_{M|A}P_{M/(A\cup\{e\})} + \sum_{B\in\mathcal{T}_e(M)} (\chi_{M|B\setminus\{e\}} - \chi_{(M|B)/e})P_{M/B}. \tag{20}$$

The flats of $M\setminus e$ and M/e partition analogously, giving

$$P^{\mathrm{rev}}_{M\setminus e}(x) = \sum_{A\not\ni e} \chi_{M|A}P_{(M/A)\setminus e} + \sum_{B\in\mathcal{T}_e(M)} \chi_{M|B\setminus\{e\}}P_{M/B},$$

$$P^{\mathrm{rev}}_{M/e}(x) = \sum_{A\in S_e(M)} \chi_{M|A}P_{M/(A\cup\{e\})} + \sum_{B\in\mathcal{T}_e(M)} \chi_{(M|B)/e}P_{M/B}.$$

The alternating sum $P^{\mathrm{rev}}_M - P^{\mathrm{rev}}_{M\setminus e} + P^{\mathrm{rev}}_{M/e}$ cancels terms corresponding to $\mathcal{T}_e(M)$, yielding

$$P^{\mathrm{rev}}_M - P^{\mathrm{rev}}_{M\setminus e} + P^{\mathrm{rev}}_{M/e} = \sum_{A\in S_e(M)\cup\{\emptyset\}} \chi_{M|A}\left(P_{M/A} - P_{(M/A)\setminus e} + xP_{(M/A)/e}\right), \tag{21}$$

where terms with $A \notin S_e(M)$ vanish because e becomes parallel to another element in M/A, forcing $P_{M/A} = P_{(M/A)\setminus e}$ and $P_{(M/A)/e} = 0$.

For every loopless minor N in which e is a non-coloop, put

$$C_e(N) = P_N - P_{N\setminus e} + xP_{N/e}.$$

Since $C_e(M)^{\mathrm{rev}} = x^{\mathrm{rk}(M)}C_e(M)(x^{-1})$, equation (21) simplifies to

$$C_e(M)^{\mathrm{rev}} - C_e(M) = \sum_{\emptyset\neq A\in S_e(M)} \chi_{M|A}(x)C_e(M/A). \tag{22}$$

Define

$$R_e(N) = \sum_{F\in S_e(N)} \tau\big(N/(F\cup\{e\})\big)x^{\mathrm{crk}_N(F)/2}P_{N|F}(x).$$

We claim, by induction on $|E(M)|$, that $C_e(M) = R_e(M)$. The defining convolution identity for P gives

$$\begin{aligned} R_e(M)^{\mathrm{rev}} &= \sum_{F\in S_e(M)} \tau\big(M/(F\cup\{e\})\big)x^{\mathrm{crk}_M(F)/2}P^{\mathrm{rev}}_{M|F}(x) \\ &= \sum_{F\in S_e(M)} \tau\big(M/(F\cup\{e\})\big)x^{\mathrm{crk}_M(F)/2} \sum_{A\in\mathcal{L}(M|F)} \chi_{M|A}(x)P_{(M|F)/A}(x) \end{aligned}$$

$$= R_e(M) + \sum_{\emptyset \neq A \in S_e(M)} \chi_{M|A}(x) R_e(M/A).$$

Indeed, in the last equality we exchange the sums. If $A \subseteq F$ with $F \in S_e(M)$, then $A \in S_e(M)$; and the flats F/A that arise are precisely the members of $S_e(M/A)$. The corresponding restrictions, contractions, coranks, and τ-coefficients agree. This is also the point at which the formula is automatically zero when e has a parallel mate: then $S_e(M) = \varnothing$.

By the induction hypothesis and (22), the difference $C_e(M) - R_e(M)$ is invariant under reversal. It remains to use the usual degree bound. Every nonempty-flat summand of $R_e(M)$ has degree strictly smaller than $\mathrm{rk}(M)/2$. The same holds for $C_e(M)$ unless $\mathrm{rk}(M)$ is even; in that case its coefficient in degree $\mathrm{rk}(M)/2$ comes only from $xP_{M/e}$ and equals

$$[x^{(\mathrm{rk}(M)-2)/2}] P_{M/e}(x) = \tau(M/e).$$

This is exactly the degree-$\mathrm{rk}(M)/2$ coefficient of the $F = \emptyset$ summand of $R_e(M)$ (and both coefficients are zero when $\emptyset \notin S_e(M)$). Hence $\deg(C_e(M) - R_e(M)) < \mathrm{rk}(M)/2$. A polynomial of degree strictly less than half its reversal degree which is invariant under reversal is zero. Thus $C_e(M) = R_e(M)$, which is the asserted formula. □

5.2. **The Chow Polynomial.**

Corollary 5.2 (Chow polynomial deletion formula)**.** *Let M be a loopless matroid and let $e \in E(M)$. Then*

$$H_M(x) = H_{M\setminus e}(x) + x \sum_{G \in \underline{S}_e(M)} H_{M|G}(x) H_{M/(G\cup\{e\})}(x). \tag{23}$$

Proof. Use the defining expression for $S_e(M)$ also when e is a coloop. Extend H by 0 to matroids with loops, with $H_\emptyset = 1$, and define

$$\kappa_N = \begin{cases} -1, & N = \emptyset, \\ \dfrac{\chi_N(x)}{x-1}, & N \neq \emptyset, \end{cases} \qquad \text{and} \qquad \lambda_N = -H_N(x).$$

The defining convolution identity for H says exactly that λ is the right convolution inverse of κ. By the characteristic-polynomial deletion identities, the hypotheses of Theorem 4.1 hold with

$$\alpha = -1, \qquad \gamma = x - 1, \qquad \varepsilon = x, \qquad \phi = 0.$$

We prove the result by induction on $|E(M)|$; the one-element case is immediate. First suppose that e is not a coloop and put $D_e(N) = H_N - H_{N\setminus e}$. If $\emptyset \in S_e(M)$, the $A = \emptyset$ summand in the inverse deletion formula cancels the exceptional term $xH_{M/e}$; if $\emptyset \notin S_e(M)$, both terms are absent. Thus Theorem 4.1 gives

$$(x-1) D_e(M) - \sum_{\substack{A \in \mathscr{L}(M) \\ e \notin A,\ A \neq \emptyset}} \chi_{M|A}(x) D_e(M/A) = x \sum_{G \in \underline{S}_e(M)} \chi_{M|G}(x) H_{M/(G\cup\{e\})}(x). \tag{24}$$

For every nonempty flat A avoiding e, the induction hypothesis applies to M/A. Reindexing the resulting sums by $G = A \cup C$, as in Lemma 2.2, gives

$$\begin{aligned}
&\sum_{\substack{A\in\mathcal{L}(M)\\ e\notin A,\ A\neq\emptyset}} \chi_{M|A} D_e(M/A) \\
&= x \sum_{G\in\underline{S}_e(M)} \left(\sum_{\substack{A\in\mathcal{L}(M|G)\\ \emptyset\neq A\subsetneq G}} \chi_{M|A} H_{(M|G)/A} \right) H_{M/(G\cup\{e\})} \\
&= x \sum_{G\in\underline{S}_e(M)} \big((x-1)H_{M|G} - \chi_{M|G}\big) H_{M/(G\cup\{e\})}.
\end{aligned}$$

The last equality is the defining convolution identity

$$(x-1)H_L = \sum_{\emptyset\neq A\in\mathcal{L}(L)} \chi_{L|A} H_{L/A} \qquad (L\neq\emptyset).$$

Substitution in (24) yields

$$(x-1)D_e(M) = x(x-1) \sum_{G\in\underline{S}_e(M)} H_{M|G} H_{M/(G\cup\{e\})},$$

and cancellation of $x-1$ proves (23).

It remains to treat a coloop e. Put $N = M/e = M\setminus e$ and $J_L = H_{L\oplus \mathsf{U}_{1,1}}$. Here $\underline{S}_e(M) = \mathcal{L}(N)\setminus\{\emptyset\}$. The case $N=\emptyset$ is immediate. For $N\neq\emptyset$, pairing the flats A and $A\cup\{e\}$ in the defining convolution for H gives

$$(x-1)J_N = \sum_{\emptyset\neq A\in\mathcal{L}(N)} \chi_{N|A} J_{N/A} + (x-1) \sum_{A\in\mathcal{L}(N)} \chi_{N|A} H_{N/A}.$$

The second sum is $(x-1)xH_N$. Applying the induction hypothesis to each $J_{N/A}$ in the first sum and reindexing once more gives, with

$$C_N = \sum_{\emptyset\neq B\in\mathcal{L}(N)} H_{N|B} H_{N/B},$$

$$\sum_{\emptyset\neq A\in\mathcal{L}(N)} \chi_{N|A} J_{N/A} = (x-1)H_N + x\big((x-1)C_N - (x-1)H_N\big).$$

Indeed, the double sum arising from the induction hypothesis is reindexed by $B = A\cup C$, and its inner sum is

$$\sum_{\substack{A\in\mathcal{L}(N|B)\\ \emptyset\neq A\subsetneq B}} \chi_{N|A} H_{(N|B)/A} = (x-1)H_{N|B} - \chi_{N|B}.$$

Consequently,

$$(x-1)J_N = (x-1)(H_N + xC_N),$$

which is precisely (23) for the coloop e. □

5.3. **The Dual Chow Polynomial.** Let $H^*_N(x)$ denote the dual Chow polynomial of a loopless matroid N, with the convention $H^*_\emptyset(x) = 1$. For the next lemma, we make the auxiliary convention that $H^*_N(x) = 0$ when N has a loop. This does not change the convolution defining G^*_M for loopless M, because contractions by flats of a loopless matroid are loopless.

Lemma 5.3 (Loop-extended dual-Chow deletion)**.** *Let N be a loopless matroid and let $p \in E(N)$ be a non-coloop. Then*

$$H^*_N(x) = H^*_{N\setminus p}(x) + H^*_{N/p}(x) + x \sum_{K \in S_p(N)} H^*_{N|K}(x) H^*_{N/(K\cup\{p\})}(x).$$

Proof. If p has no parallel mate, the deletion formula of Caiolo–Ferroni–Hoster has the form

$$H^*_N = H^*_{N\setminus p} + (1+x)H^*_{N/p} + x \sum_{K \in \underline{S}_p(N)} H^*_{N|K} H^*_{N/(K\cup\{p\})}.$$

In this case $\emptyset \in S_p(N)$, and the $K = \emptyset$ summand in the displayed formula of the lemma is $xH^*_{N/p}$, because $H^*_{N|\emptyset} = H^*_\emptyset = 1$. Thus the two forms agree.

Suppose instead that p has a parallel mate q. Then $S_p(N) = \varnothing$: a flat avoiding p also avoids q, whereas adjoining p without q cannot produce a flat. Deleting p preserves the ranked lattice of flats, so $H^*_N = H^*_{N\setminus p}$. On the other hand, q is a loop of N/p, and the auxiliary zero-on-loops convention gives $H^*_{N/p} = 0$. This proves the stated recurrence in every case. □

5.4. **The Z-Polynomial.** The Z-polynomial of M is defined by the convolution

$$Z_M(x) = (\zeta^{\mathrm{rev}} * P)_M = \sum_{A \in \mathcal{L}(M)} x^{\mathrm{rk}(A)} P_{M/A}(x), \tag{25}$$

where $\zeta^{\mathrm{rev}}_N(x) = x^{\mathrm{rk}(N)}$. The following is a direct specialization of Corollary 3.2.

Corollary 5.4 (Z-polynomial deletion formula)**.** *Let M be loopless and let $e \in E(M)$ be a non-coloop. Then*

$$Z_M(x) = Z_{M\setminus e}(x) + \sum_{F \in S_e(M)} \tau\big(M/(F\cup e)\big) x^{\mathrm{crk}_M(F)/2} Z_{M|F}(x).$$

Proof. Extend P by 0 to matroids with loops, and put

$$\phi(Q) = \begin{cases} \tau(Q) x^{(\mathrm{rk}(Q)+1)/2}, & \text{if } Q \text{ is loopless and } \mathrm{rk}(Q) \text{ is odd,} \\ 0, & \text{otherwise.} \end{cases}$$

Apply Corollary 3.2 with

$$g_N = x^{\mathrm{rk}(N)}, \qquad h_N = P_N(x).$$

The Kazhdan–Lusztig deletion formula gives

$$\alpha_h = -x, \qquad \Phi_h = \phi, \qquad \Psi_h = 0,$$

while $g_N = g_{N\setminus p}$ for a non-coloop p and $g_N = x g_{N/p}$ for a coloop p, so that

$$\alpha_g = 0, \qquad \gamma_g = x, \qquad \Phi_g = \Psi_g = 0.$$

The convolution in the Convolution Deletion Formula is $f_N = Z_N(x)$, and its additional loop-vanishing hypotheses hold. Since $\delta = -x + x = 0$, only the Φ_h-term remains. Finally, for $F \in S_e(M)$,

$$\phi\big(M/(F\cup\{e\})\big) = \tau\big(M/(F\cup e)\big) x^{\mathrm{crk}_M(F)/2},$$

because $\mathrm{rk}(M/(F\cup\{e\})) + 1 = \mathrm{crk}_M(F)$. This proves the formula. □

Lemma 5.5. *For any loopless matroid M and $e \in E(M)$ that is not a coloop, the reversed Kazhdan–Lusztig polynomial satisfies:*

$$P^{\mathrm{rev}}_M(x) = P^{\mathrm{rev}}_{M\setminus e}(x) - P^{\mathrm{rev}}_{M/e}(x) + \sum_{F \in S_e(M)} \tau(M/(F\cup\{e\})) x^{\mathrm{crk}_M(F)/2} P^{\mathrm{rev}}_{M|F}(x)$$

Proof. The standard deletion formula for the Kazhdan-Lusztig polynomial $P_M(x)$ is:

$$P_M(x) = P_{M\setminus e}(x) - xP_{M/e}(x) + \sum_{F\in S_e(M)} \tau(M/(F\cup e))x^{\mathrm{crk}_M(F)/2}P_{M|F}(x).$$

Substituting x with x^{-1} and multiplying by $x^{\mathrm{rk}(M)}$ yields the stated formula for $P^{\mathrm{rev}}_M(x)$. □

5.5. **The Right Augmented Chow Polynomial.**

$$F_N(x) = (H * P^{\mathrm{rev}})_N = \sum_{A\in\mathcal{L}(N)} H_{N|A}(x)P^{\mathrm{rev}}_{N/A}(x)$$

for every loopless matroid N. For the bookkeeping required by the Convolution Deletion Formula, extend H by 0 to matroids with loops.

Corollary 5.6 (Right augmented Chow deletion formula)**.** *Let M be a simple matroid and let $e \in E(M)$ be a non-coloop. The right augmented Chow polynomial satisfies:*

$$\begin{aligned} F_M(x) = F_{M\setminus e}(x) &+ x\sum_{G\in\underline{S}_e(M)} H_{M|G}(x)F_{M/(G\cup e)}(x) \\ &+ \sum_{F\in S_e(M)} \tau(M/(F\cup e))x^{\mathrm{crk}_M(F)/2}F_{M|F}(x). \end{aligned}$$

Proof. Extend P^{rev} by 0 to matroids with loops, and set

$$\phi(Q) = \begin{cases} \tau(Q)x^{(\mathrm{rk}(Q)+1)/2}, & \text{if } Q \text{ is loopless and } \mathrm{rk}(Q) \text{ is odd,} \\ 0, & \text{otherwise.} \end{cases}$$

We apply Corollary 3.2 with

$$g_N = H_N(x), \qquad h_N = P^{\mathrm{rev}}_N(x).$$

Corollary 5.2 gives the required g-recurrences with

$$\alpha_g = 0, \qquad \gamma_g = 1, \qquad \Psi_g = 0,$$

and

$$\Phi_g(Q) = \begin{cases} xH_Q(x), & \text{if } E(Q) \neq \varnothing, \\ 0, & \text{if } Q = \emptyset. \end{cases}$$

Indeed, the non-coloop formula has precisely the $\underline{S}_p(N)$ sum, while the coloop formula has the extra term $H_{N/p}$ required by the Convolution Deletion Formula. Lemma 5.5 supplies the h-recurrence with

$$\alpha_h = -1, \qquad \Phi_h = \phi, \qquad \Psi_h = 0.$$

The equality

$$\mathrm{rk}\big(N/(K\cup\{p\})\big) + 1 = \mathrm{crk}_N(K)$$

for $K \in S_p(N)$ identifies its Φ_h-weight with the boundary coefficient in that lemma.

The convolution in the Convolution Deletion Formula is $f_N = F_N$, and h vanishes on looped matroids. The remaining hypotheses of Corollary 3.2 hold because $\Psi_h = 0$ and $\Phi_g(\emptyset) = 0$. Moreover,

$$\delta = \alpha_h + \gamma_g - \alpha_g = -1 + 1 = 0.$$

Consequently, the Φ_h-sum in the loop-vanishing corollary is

$$\sum_{F\in S_e(M)} \tau\big(M/(F\cup e)\big)x^{\mathrm{crk}_M(F)/2}F_{M|F}(x),$$

whereas its Φ_g-sum is

$$x \sum_{G \in \underline{S}_e(M)} H_{M|G}(x) F_{M/(G \cup e)}(x).$$

These are exactly the two terms in the stated deletion formula. □

5.6. **The Left Augmented Dual Chow Polynomial.** The left augmented dual Chow polynomial is defined by

$$G^*_M(x) = \sum_{F \in \mathcal{L}(M)} Q^{\mathrm{rev}}_{M|F}(x) H^*_{M/F}(x),$$

where $H^*_M(x)$ is the dual Chow polynomial and $Q^{\mathrm{rev}}_M(x) = x^{\mathrm{rk}(M)} Q_M(x^{-1})$ is the reversed inverse Kazhdan–Lusztig polynomial.

Corollary 5.7 (Left augmented dual Chow deletion formula)**.** *Let M be a simple matroid, and let $e \in E(M)$ be a non-coloop. Then The left augmented dual Chow polynomial $G^*_M(x)$ satisfies the deletion formula:*

$$G^*_M(x) = G^*_{M\setminus e}(x) + (1+x) G^*_{M/e}(x) + x \sum_{F \in S_e(M)} G^*_{M|F}(x) H^*_{M/(F \cup e)}(x) - \sum_{\substack{K \in \mathcal{T}_e(M) \\ \mathrm{rk}_M(K) \text{ even}}} \tau\big((M|K)/e\big) x^{\mathrm{rk}_M(K)/2} G^*_{M/K}(x).$$

Proof. Put $q_N = Q^{\mathrm{rev}}_N(x)$, using the zero-on-loops convention already fixed for Q, and define

$$\psi(Q) = \begin{cases} -\tau(Q) x^{(\mathrm{rk}(Q)+1)/2}, & \text{if } Q \text{ is loopless and } \mathrm{rk}(Q) \text{ is odd,} \\ 0, & \text{otherwise.} \end{cases}$$

The inverse Kazhdan–Lusztig deletion formula, proved below as Corollary 5.9 gives, after reversal, for a loopless pointed matroid (N, p) with p a non-coloop,

$$q_N = q_{N\setminus p} + (1+x) q_{N/p} + \sum_{K \in \mathcal{T}_p(N)} \psi\big((N|K)/p\big) q_{N/K}.$$

If p is a coloop, then $Q_N = Q_{N/p}$, and reversal gives

$$q_N = x q_{N/p}.$$

We may therefore apply Corollary 3.2 with

$$h_N = H^*_N(x), \qquad g_N = q_N,$$

and parameters

$$\alpha_h = 1, \qquad \Phi_h(Q) = x H^*_Q(x), \qquad \Psi_h = 0,$$
$$\alpha_g = 1 + x, \qquad \gamma_g = x, \qquad \Phi_g = 0, \qquad \Psi_g(Q) = \psi(Q).$$

Lemma 5.3 supplies the h-recurrence, including the case in which p has a parallel mate. Its auxiliary extension also gives the required loop-vanishing condition. The convolution in the Convolution Deletion Formula is

$$f_N = \sum_{A \in \mathcal{L}(N)} Q^{\mathrm{rev}}_{N|A}(x) H^*_{N/A}(x) = G^*_N(x),$$

and

$$\delta = \alpha_h + \gamma_g - \alpha_g = 1 + x - (1+x) = 0.$$

Thus the Φ_h-term in the loop-vanishing corollary is

$$x \sum_{F \in S_e(M)} G^*_{M|F}(x) H^*_{M/(F \cup e)}(x),$$

and the Ψ_g-term is

$$\sum_{K\in\mathcal{T}_e(M)} \psi\big((M|K)/e\big)G^*_{M/K}(x).$$

Since

$$\mathrm{rk}\big((M|K)/e\big) = \mathrm{rk}_M(K) - 1,$$

the latter sum is exactly

$$-\sum_{\substack{K\in\mathcal{T}_e(M)\\ \mathrm{rk}_M(K)\text{ even}}} \tau\big((M|K)/e\big)x^{\mathrm{rk}_M(K)/2}G^*_{M/K}(x).$$

Together with the deletion and contraction terms of the master corollary, this is the displayed formula. □

Remark 5.8 (The G^* contraction coefficient)**.** For the right augmented Chow polynomial, the nonzero master boundary weight is Φ_g, so its extra contribution is indexed by $S_e(M)$. Here $\Phi_g = 0$; instead, the G^* specialization has the contraction coefficient

$$\Psi_g(Q) = \psi(Q).$$

Consequently the Convolution Deletion Formula evaluates it on the contraction $(M|K)/e$ and produces

$$\Psi_g\big((M|K)/e\big) = \psi\big((M|K)/e\big).$$

Thus the contraction by e is taken *after* restricting to K, which is the source of the $\mathcal{T}_e(M)$-sum in the G^* formula.

5.7. **The Inverse Kazhdan-Lusztig Polynomial.** For a loopless matroid M, put

$$\widehat{Q}_M(x) = (-1)^{\mathrm{rk}(M)}Q_M(x).$$

Extend P, Q, and τ by zero to matroids with loops. With this convention, $\widehat{Q}$ is the right convolution inverse of P:

$$\sum_{A\in\mathcal{L}(M)} P_{M|A}(x)\widehat{Q}_{M/A}(x) = \delta_{\emptyset,M}. \tag{26}$$

For looped M, this identity is automatic, because every flat contains a loop and every restricted P-factor therefore vanishes. Corollary 4.2 therefore gives the deletion formula formally from the deletion formula for P.

Corollary 5.9 (Inverse Kazhdan–Lusztig deletion formula)**.** *Let M be a loopless matroid and let $e \in E(M)$ be a non-coloop. Then*

$$Q_M(x) = Q_{M\setminus e}(x) + (1+x)Q_{M/e}(x) - \sum_{\substack{F\in\mathcal{T}_e(M)\\ \mathrm{rk}(F)\ \textit{even}}} \tau\big((M|F)/e\big)x^{\mathrm{rk}(F)/2}Q_{M/F}(x). \tag{27}$$

Proof. Since $P_\emptyset = 1$, Corollary 4.2 applies with

$$\kappa_N = P_N(x), \qquad \lambda_N = \widehat{Q}_N(x), \qquad \alpha = -x, \qquad \gamma = 1,$$

and define

$$\phi(N) = \begin{cases} \tau(N)x^{(\mathrm{rk}(N)+1)/2}, & \text{if } N \text{ is loopless and } \mathrm{rk}(N) \text{ is odd,}\\ 0, & \text{otherwise.}\end{cases}$$

Indeed, the Kazhdan–Lusztig deletion recurrence has the form

$$P_N = P_{N\setminus p} - xP_{N/p} + \sum_{G\in S_p(N)} \phi\big(N/(G\cup\{p\})\big)P_{N|G},$$

because

$$\mathrm{rk}\big(N/(G\cup\{p\})\big)+1=\mathrm{crk}_N(G).$$

Its coloop relation is $P_N = P_{N/p}$. Thus the corollary gives

$$\widehat{Q}_M=\widehat{Q}_{M\setminus e}-(1+x)\widehat{Q}_{M/e}-\sum_{\substack{F\in\mathcal{T}_e(M)\\ \mathrm{rk}(F)\text{ even}}}\tau\big((M|F)/e\big)x^{\mathrm{rk}(F)/2}\widehat{Q}_{M/F}.$$

The rank of $((M|F)/e)$ is $\mathrm{rk}(F)-1$, so only even-rank F can contribute. If $r=\mathrm{rk}(M)$, then

$$\mathrm{rk}(M\setminus e)=r,\qquad \mathrm{rk}(M/e)=r-1,\qquad \mathrm{rk}(M/F)=r-\mathrm{rk}(F).$$

Replacing $\widehat{Q}_N$ by $(-1)^{\mathrm{rk}(N)}Q_N$ therefore reverses the contraction sign and preserves every boundary sign, proving (27). □

5.8. **The Inverse Z-Polynomial.** For a loopless matroid M, put

$$\widehat{Y}_M(x)=(-1)^{\mathrm{rk}(M)}Y_M(x).$$

For the following convolutional argument, extend Z, $\widehat{Y}$ (and hence Y), and τ by zero to matroids with loops. This is an auxiliary bookkeeping convention; it does not evaluate (25) on a looped matroid. With this convention, $\widehat{Y}$ is the right convolution inverse of Z:

$$\sum_{A\in\mathcal{L}(M)}Z_{M|A}(x)\widehat{Y}_{M/A}(x)=\delta_{\emptyset,M}. \tag{28}$$

For looped M, this identity is automatic, because every flat contains a loop and every restricted Z-factor therefore vanishes.

Corollary 5.10 (Inverse Z-polynomial deletion formula)**.** *Let M be a loopless matroid and let $e\in E(M)$ be a non-coloop. Then*

$$Y_M(x)=Y_{M\setminus e}(x)+(1+x)Y_{M/e}(x)-\sum_{\substack{F\in\mathcal{T}_e(M)\\ \mathrm{rk}(F)\ \textit{even}}}\tau\big((M|F)/e\big)x^{\mathrm{rk}(F)/2}Y_{M/F}(x). \tag{29}$$

Proof. Since $Z_\emptyset=1$, Corollary 4.2 applies with

$$\kappa_N=Z_N(x),\qquad \lambda_N=\widehat{Y}_N(x),\qquad \alpha=0,\qquad \gamma=1+x,$$

and

$$\phi(N)=\begin{cases}\tau(N)x^{(\mathrm{rk}(N)+1)/2}, & \text{if } N \text{ is loopless and } \mathrm{rk}(N) \text{ is odd},\\ 0, & \text{otherwise.}\end{cases}$$

The loopless deletion formula of Corollary 5.4 has the required form with $\alpha=0$, since

$$\mathrm{rk}\big(N/(G\cup\{p\})\big)+1=\mathrm{crk}_N(G).$$

If p is a coloop of N, the flats of N occur in pairs A and $A\cup\{p\}$ with $A\in\mathcal{L}(N/p)$. Adjoining a coloop does not change the Kazhdan–Lusztig polynomial, so the corresponding summands in (25) are 1 and x times the same summand of $Z_{N/p}$. Hence

$$Z_N=(1+x)Z_{N/p}.$$

Thus $\alpha-\gamma=-(1+x)$, and the corollary gives

$$\widehat{Y}_M=\widehat{Y}_{M\setminus e}-(1+x)\widehat{Y}_{M/e}-\sum_{\substack{F\in\mathcal{T}_e(M)\\ \mathrm{rk}(F)\text{ even}}}\tau\big((M|F)/e\big)x^{\mathrm{rk}(F)/2}\widehat{Y}_{M/F}.$$

Again, a nonzero boundary term has $\mathrm{rk}(F)$ even. If $r=\mathrm{rk}(M)$, then

$$\mathrm{rk}(M\setminus e)=r,\qquad \mathrm{rk}(M/e)=r-1,\qquad \mathrm{rk}(M/F)=r-\mathrm{rk}(F).$$

Replacing $\widehat{Y}_N$ by $(-1)^{\mathrm{rk}(N)}Y_N$ therefore reverses the contraction sign and preserves every boundary sign, proving (29). □

Remark 5.11 (Why the inverse formulas coincide)**.** The deletion formulas in Corollaries 5.9 and 5.10 coincide because the forward kernels have the same boundary weight ϕ, and

$$(-x) - 1 = -(1+x) = 0 - (1+x).$$

Equivalently, the pairs $(\alpha, \gamma) = (-x, 1)$ for P and $(0, 1+x)$ for Z have the same difference $\alpha - \gamma$. The loop-vanishing inverse formula depends on the forward recurrence precisely through this difference and ϕ; after the same rank-sign normalization, it therefore yields identical recurrences for Q and Y.

6. Remark on Gamma-Positivity

Remark 6.1. Recall that a symmetric polynomial $p(x)$ of degree d is γ-positive if it is a nonnegative linear combination of the polynomials $x^i(1+x)^{d-2i}$. The deletion formula for the right augmented Chow polynomial is compatible with its symmetry, but it does not by itself imply γ-positivity: its summands have different symmetry degrees, and the formula includes the coefficients τ of Kazhdan–Lusztig polynomials. Establishing γ-positivity therefore requires an additional structural argument, which is not supplied here.

7. Examples and Specializations

We illustrate the deletion formula for the right augmented Chow polynomial in several matroid classes. The following calculations specialize the formula proved above. We write

$$\mathsf{B}_j \coloneqq \mathsf{U}_{j,j}, \qquad \mathsf{C}_j \coloneqq \mathsf{U}_{j-1,j},$$

for the Boolean matroid of rank j and the circuit of size j, respectively. We set

$$A_j(x) \coloneqq H_{\mathsf{B}_j}(x), \qquad \widetilde{A}_j(x) \coloneqq F_{\mathsf{B}_j}(x),$$

which denote the Chow and right augmented Chow polynomials of the Boolean matroid of rank j. The class-specific calculations in the subsequent sections specialize Corollary 5.6 for a simple matroid M:

$$\begin{aligned} F_M(x) = F_{M\setminus e}(x) + x \sum_{G \in \underline{S}_e(M)} & H_{M|G}(x) F_{M/(G\cup\{e\})}(x) \\ + \sum_{G \in S_e(M)} & \tau(M/(G\cup\{e\}))\, x^{\mathrm{crk}_M(G)/2} F_{M|G}(x). \end{aligned} \tag{D_F}$$

Whenever a Kazhdan–Lusztig τ-coefficient $\tau(N)$ is nonzero, $\mathrm{rk}(N)$ is odd; hence the corresponding half-integral exponent in (D_F) is an integer.

7.1. Uniform Matroids.

Theorem 7.1. *Let $2 \le r < n$, let $M = \mathsf{U}_{r,n}$, and let $e \in E(M)$. Then*

$$S_e(M) = \{G \subseteq E(M) \setminus \{e\} : |G| \le r-2\}.$$

Consequently, the augmented Chow polynomial satisfies

$$\begin{aligned} F_{\mathsf{U}_{r,n}}(x) = F_{\mathsf{U}_{r,n-1}}(x) + x \sum_{j=1}^{r-2} & \binom{n-1}{j} A_j(x) F_{\mathsf{U}_{r-j-1,n-j-1}}(x) \\ + \sum_{j=0}^{r-2} & \binom{n-1}{j} \tau\big(\mathsf{U}_{r-j-1,n-j-1}\big)\, x^{(r-j)/2} \widetilde{A}_j(x). \end{aligned} \tag{30}$$

The first sum is empty when $r = 2$.

Proof. The proper flats of $\mathsf{U}_{r,n}$ are precisely the subsets of cardinality strictly less than r. Hence, for a flat G not containing e, $G \cup \{e\} \in \mathscr{L}(\mathsf{U}_{r,n})$ if and only if $|G| + 1 < r$. Therefore,

$$G \in S_e(M) \iff |G| \leq r - 2.$$

If $|G| = j \leq r - 2$, then G is independent, so $M|G \cong \mathsf{B}_j$ and $M/(G \cup \{e\}) \cong \mathsf{U}_{r-j-1,n-j-1}$. Moreover, $\operatorname{crk}_M(G) = r - j$. There are exactly $\binom{n-1}{j}$ such flats. Thus, the quadratic term in (D_F) becomes

$$x \sum_{j=1}^{r-2} \binom{n-1}{j} A_j(x) F_{\mathsf{U}_{r-j-1,n-j-1}}(x),$$

while the τ-term becomes

$$\sum_{j=0}^{r-2} \binom{n-1}{j} \tau\big(\mathsf{U}_{r-j-1,n-j-1}\big)\, x^{(r-j)/2} \widetilde{A}_j(x).$$

Finally, $M \setminus e \cong \mathsf{U}_{r,n-1}$, which establishes (30). □

Remark 7.2 (Boundary Cases)**.** For $r = 2$, we have $F_{\mathsf{U}_{2,n}}(x) = F_{\mathsf{U}_{2,n-1}}(x) + x$, because $\tau(\mathsf{U}_{1,n-1}) = 1$. The case $r = n$ is excluded because every element is a coloop. The case $r = 1 < n$ is excluded from the stated simple-matroid theorem because $\mathsf{U}_{1,n}$ is not simple.

7.2. **Glued Cycles.** Let $\mathsf{C}_{a,b}$ be the graphic matroid of two cycles of lengths a and b, with $a, b \geq 3$, glued along a common edge e. Put $p \coloneqq a - 1$ and $q \coloneqq b - 1$. The graph consists of e and two internally disjoint paths of lengths p and q between the endpoints of e.

Lemma 7.3. *A subset $G \subseteq E(\mathsf{C}_{a,b}) \setminus \{e\}$ lies in $S_e(\mathsf{C}_{a,b})$ if and only if G omits at least two edges from each of the two paths.*

Proof. Let P_a and P_b denote the two paths. If G omits only one edge f from P_a, then in $G \cup \{e\}$ the endpoints of f are connected by the remaining edges of P_a together with e. Thus $f \in \operatorname{cl}(G \cup \{e\})$, so $G \cup \{e\}$ is not a flat. The same applies to P_b. Conversely, suppose at least two edges are omitted from each path. Then neither G nor $G \cup \{e\}$ contains all but one edge of any cycle of the theta graph. Hence, no omitted edge has both endpoints in the same connected component of the corresponding spanning subgraph. Therefore, both G and $G \cup \{e\}$ are flats. □

Lemma 7.4. *Suppose $G \in S_e(\mathsf{C}_{a,b})$ omits s edges from P_a and t edges from P_b. Then $s \geq 2$, $t \geq 2$, and*

$$\mathsf{C}_{a,b}|G \cong \mathsf{B}_{p+q-s-t}, \qquad \mathsf{C}_{a,b}/(G \cup \{e\}) \cong \mathsf{C}_s \oplus \mathsf{C}_t.$$

Proof. The restriction to G is a forest, so every element of G is a coloop in the restriction. Hence it is Boolean. After contracting e and all selected edges of each path, the s uncontracted edges of the first path form a cycle, as do the t uncontracted edges of the second path. These two cycles meet only at a vertex, so their cycle matroids form a direct sum. □

Lemma 7.5. *For $s, t \geq 2$, $\tau(\mathsf{C}_s \oplus \mathsf{C}_t) = 0$.*

Proof. The Kazhdan–Lusztig polynomial is multiplicative under direct sums: $P_{\mathsf{C}_s \oplus \mathsf{C}_t}(x) = P_{\mathsf{C}_s}(x) P_{\mathsf{C}_t}(x)$. If $(s - 1) + (t - 1)$ is even, the assertion follows from the definition of τ. If it is odd, precisely one of $s - 1$ and $t - 1$ is even. The degree bounds $\deg P_{\mathsf{C}_s}(x) < \frac{s-1}{2}$ and $\deg P_{\mathsf{C}_t}(x) < \frac{t-1}{2}$ imply $\deg P_{\mathsf{C}_s \oplus \mathsf{C}_t}(x) < \frac{s+t-3}{2}$, so the coefficient defining τ vanishes. □

Theorem 7.6. *The deletion formula for the common edge is purely quadratic:*

$$F_{\mathsf{C}_{a,b}}(x) = F_{\mathsf{C}_{a+b-2}}(x) + x \sum_{\substack{s=2}}^{a-1} \sum_{\substack{t=2 \\ (s,t)\neq(a-1,b-1)}}^{b-1} \binom{a-1}{s}\binom{b-1}{t} A_{a+b-2-s-t}(x) F_{\mathsf{C}_s \oplus \mathsf{C}_t}(x). \tag{31}$$

Proof. Deleting the common edge joins the two paths into a single cycle of length $(a-1)+(b-1) = a+b-2$. Thus $\mathsf{C}_{a,b} \setminus e \cong \mathsf{C}_{a+b-2}$. For fixed s and t, the omitted edges can be chosen in $\binom{a-1}{s}\binom{b-1}{t}$ ways. Lemma 7.4 identifies the restriction and contraction appearing in (D_F). The pair $(s,t) = (a-1, b-1)$ corresponds to $G = \varnothing$, which must be omitted from the quadratic sum. Lemma 7.5 dictates that the entire τ-sum identically vanishes. □

7.3. **Projective Geometry Deletions.** Let $M = \mathrm{PG}(r-1, q)$ with $r \geq 2$, and let e be a point.

Lemma 7.7. *For a projective geometry, $S_e(M) = \{\varnothing\}$.*

Proof. Both $\varnothing$ and $\{e\}$ are flats, so $\varnothing \in S_e(M)$. Suppose $G \neq \varnothing$ is a flat not containing e. Choose a point $f \in G$. Since G is a projective subspace not containing e, its intersection with the projective line $\langle e, f\rangle$ is exactly $\{f\}$. The line has at least three points, so $\langle e, f\rangle \setminus \{e, f\} \neq \varnothing$. All these additional points lie in the closure of $G \cup \{e\}$, but not in $G \cup \{e\}$. Therefore, $G \cup \{e\}$ is not a flat. □

Lemma 7.8. *For $M = \mathrm{PG}(r-1, q)$,*

$$\tau(M/e) = \begin{cases} 1, & r = 2, \\ 0, & r \geq 3. \end{cases}$$

Proof. Projective geometries are modular, meaning $P_{M/e}(x) = 1$. If $r = 2$, M/e has rank 1, and consequently $\tau(M/e) = [x^0]1 = 1$. If $r \geq 3$, either $r-1$ is even (so τ vanishes by definition), or $r-1$ is odd and the coefficient defining τ has positive degree in the constant polynomial 1, yielding zero. □

Theorem 7.9. *A single-point deletion satisfies*

$$F_{\mathrm{PG}(r-1,q)\setminus e}(x) = F_{\mathrm{PG}(r-1,q)}(x) - \mathbf{1}_{\{r=2\}} x. \tag{32}$$

In particular, $F_{\mathrm{PG}(r-1,q)\setminus e}(x) = F_{\mathrm{PG}(r-1,q)}(x)$ for every $r \geq 3$.

Proof. Since $\underline{S}_e(M) = \varnothing$, the quadratic sum in ($\mathrm{D}_F$) is empty. The only term in the τ-sum is $G = \varnothing$, for which $\mathrm{crk}_M(\varnothing) = r$ and $F_{M|\varnothing}(x) = 1$. Thus $F_M(x) = F_{M\setminus e}(x) + \tau(M/e)x^{r/2}$, and Lemma 7.8 yields the result. □